

SAROP Workspace

Prototype Walkthrough

A module-by-module tour of the interactive prototype: the purpose and features of each of the nine surfaces, with a screen from each.

Draft v0.1

Internal . Pre-gate . Desirability prototype

Built on the SAROP Workspace design system and the ten-module product definition suite

How to read this booklet

What this is. A walkthrough of the SAROP Workspace interactive prototype, the single-operation product in the TacEdge SAR line. Each of the nine modules gets a page: a screen from the working prototype, what the surface is for, and what is on it. The screens are live and interactive; the images here are captured from them.

One product, not nine mockups. Every surface is built on one design system and reads and writes to one operational record, the incident. That is why the same operation threads through all nine screens: the stand-up decision, the task, the field captures, the forms and the reconciliation are the same record seen from different surfaces. Land and marine are a configuration, shown by the Land / Marine toggle on almost every screen, not two separate products.

What it tests, and what it does not. These screens are a desirability prototype. They exist to answer one question with a real SAR coordinator and coordinating authority: is the coordinator-first design right, and are the flows fast and legible enough to be used on a live job. They do not test offline field reliability, which is the load-bearing engineering risk and can only be proven by the working build. Where that distinction matters most, on the field capture surface, the prototype says so on the screen.

The flows. Five flows carry the product, and the booklet is ordered to walk them: stand up an operation (F1), see the picture (F3), task a team (F2), capture from the field (F4), and close out to the confirmed record (F5). The remaining surfaces, the log, the forms, callout and resources, support and connect them.

M1 Stand-up & Incident Control

Flow F1 · cold stand-up Primary user: SAR Coordinator Target: running operation under 10 minutes

TacEdge SAR SAROP WORKSPACE DOMAIN LAND MARINE 00:01 ELAPSED · TARGET < 10:00

Stand up an operation

From notification to a running operation. Fill what you know, appoint yourself or a controller, and go active. Everything else can follow.

NEW OPERATION

DOMAIN

● Land ● Marine (close-to-shore)

Sets the base map, urgency scoring, forms and terminology for the whole operation.

CATEGORY

Category I Local coordination. Cat I / II is a marker, not the config axis.

REFERENCE

SAR-2607-042 AUTO-ASSIGNED

DESCRIPTION

One line on the incident, e.g. Overdue tramper, Kepler Track, last seen Luxmore Hut.

POLICE EVENT NUMBER - optional

P-number or RCCNZ reference, if known

The CAD and records link. Can be added later.

INCIDENT CONTROLLER

Name of the controller (you, if standing up alone)

The one role needed to go active. Add the rest of the IMT as people arrive.

+ Intelligence + Planning + Operations + Logistics

INCIDENT Investigating

SAR-2607-042

DOMAIN LAND

CATEGORY Category I

DESCRIPTION Not set

POLICE REF Not set

CONTROLLER Not appointed

🔒 Locked - - - - - unassigned

STAND-UP

The five steps to a running operation.

☒ Reached without admin no specialist setup

☐ Created & described

☒ Domain & category set

☐ Controller appointed

☐ Active

M1 — standing up an operation, with the sub-ten-minute stopwatch running.

Purpose

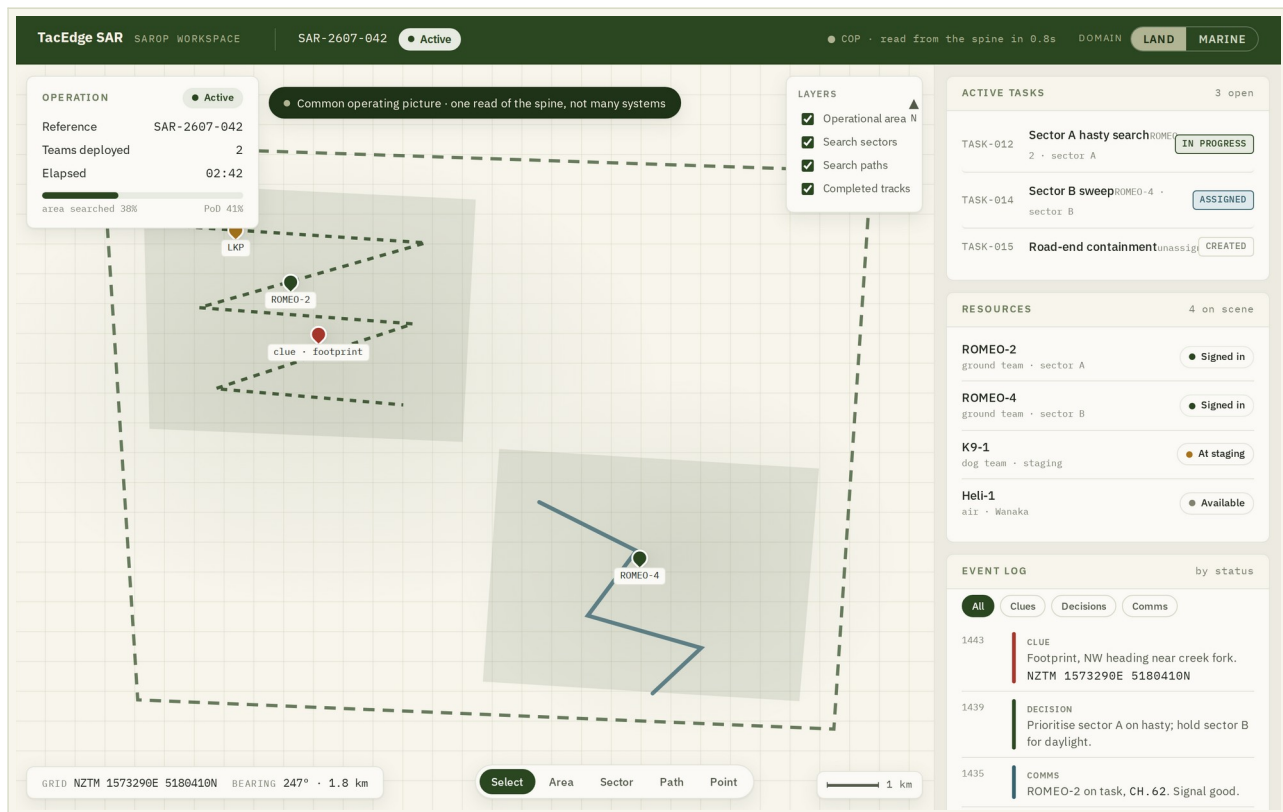
The cold-start surface. A notification arrives and an on-call, possibly skill-faded coordinator has to get from nothing to a running operation, fast and without specialist admin. M1 is that first surface, and it is deliberately spare: only what is needed to go active, with everything else able to follow.

On the screen

- **Stand-up stopwatch.** A timer runs from the moment stand-up opens, framed against the under-ten-minute target, and turns to caution if it is crossed. The metric is on the screen, not in a manual.
- **The minimum path.** Domain first (it configures the whole operation), then an auto-assigned reference, a one-line description, an optional Police event number, and the Incident Controller. Only the description and controller are needed to go active.
- **The operation, assembling live.** A summary card builds as you type, carrying the status pill and the locked attribution stamp, alongside a five-step checklist that ticks off as you reach each condition.
- **Go active.** One action flips the incident from Investigating to Active, stamps the record, and hands off to the common operating picture.

M2 The COP

Flow F3 · picture on login Read by the whole IMT Target: rendered in under 10 seconds



M2 — the common operating picture: one read of the record, not an assembly from many systems.

Purpose

The common operating picture the whole incident management team reads on login. Its job is to show the entire operation at a glance and orient anyone in seconds, and to be one read of the single record rather than an assembly stitched together from separate systems, which was the coordination failure the product exists to fix.

On the screen

- **Map-dominant.** The operational area, search sectors, planned search paths, completed tracks and feature pins (last known point, clues, teams) sit on a base map that the domain toggle switches between land and marine.
- **The operation, floating on the map.** An operation and progress panel (area searched, probability of detection), a layer switcher, a live grid-reference readout, a scale bar and a plan toolbar to define areas, sectors, paths and points without leaving the surface.
- **The record in the rail.** Active tasks with their status, resources with their sign-in state, and the event log filtered by status, each fact set in the mono data language.
- **Cross-reference.** Selecting a task highlights its sector on the map and its team; selecting a resource or sector does the reverse. Everything is one connected record.

M3 Tasking

Flow F2 · task a team Primary user: SAR Coordinator Target: task issued under 1 minute

The screenshot displays the TacEdge SAR Tasking interface. At the top, a dark green header bar contains the text 'TacEdge SAR SAROP WORKSPACE', 'SAR-2687-042', and an 'Active' status indicator. On the right, it shows a timer '00:00 task · target < 1:00' and domain tabs for 'LAND' and 'MARINE'. The main content area is titled 'Tasking' with a subtitle 'Create it, assign it, attach where, push it to the field. One task object, whether you task direct or for-affect.' The interface is divided into two main panels. The left panel, titled 'NEW TASK' (TASK-016), contains several sections: 'SUBJECT' with a text input field containing 'e.g. Sector C creek-line search'; 'TASK TYPE' with buttons for 'Hasty search', 'Sector sweep', 'Containment', and 'Track / clue'; 'ASSIGN RESOURCE' with three resource selection buttons: 'ROMEO-2 ground · sector A', 'ROMEO-4 ground · sector B', and 'K9-1 dog · staging'; a note 'One or more. Assigned assets and their history stay linked to the task.'; 'LOCATION' with buttons for 'Search area', 'Search path', and 'Single location', and a selected option '✓ Sector C attached · 0.82 km²'; and 'ROUTING' with a 'LAND · DIRECT TASKING' section containing a button 'Direct task to the assigned team' and a note 'The team works under your immediate control. You task and track them directly (TASK-10)'. The right panel, titled 'TASK BOARD' (3 open), shows a list of tasks with filters 'All', 'Created', 'Assigned', and 'In progress'. It lists three tasks: 'TASK-012 Sector A hasty search' (IN PROGRESS), 'TASK-014 Sector B sweep' (ASSIGNED), and 'TASK-015 Road-end containment' (CREATED). At the bottom of the 'NEW TASK' panel, there are buttons for 'App', 'SMS', 'Email', and 'Push to field', along with a note 'Add a subject and a resource to push.'

M3 — one task object, with the routing block set to land (direct) tasking.

Purpose

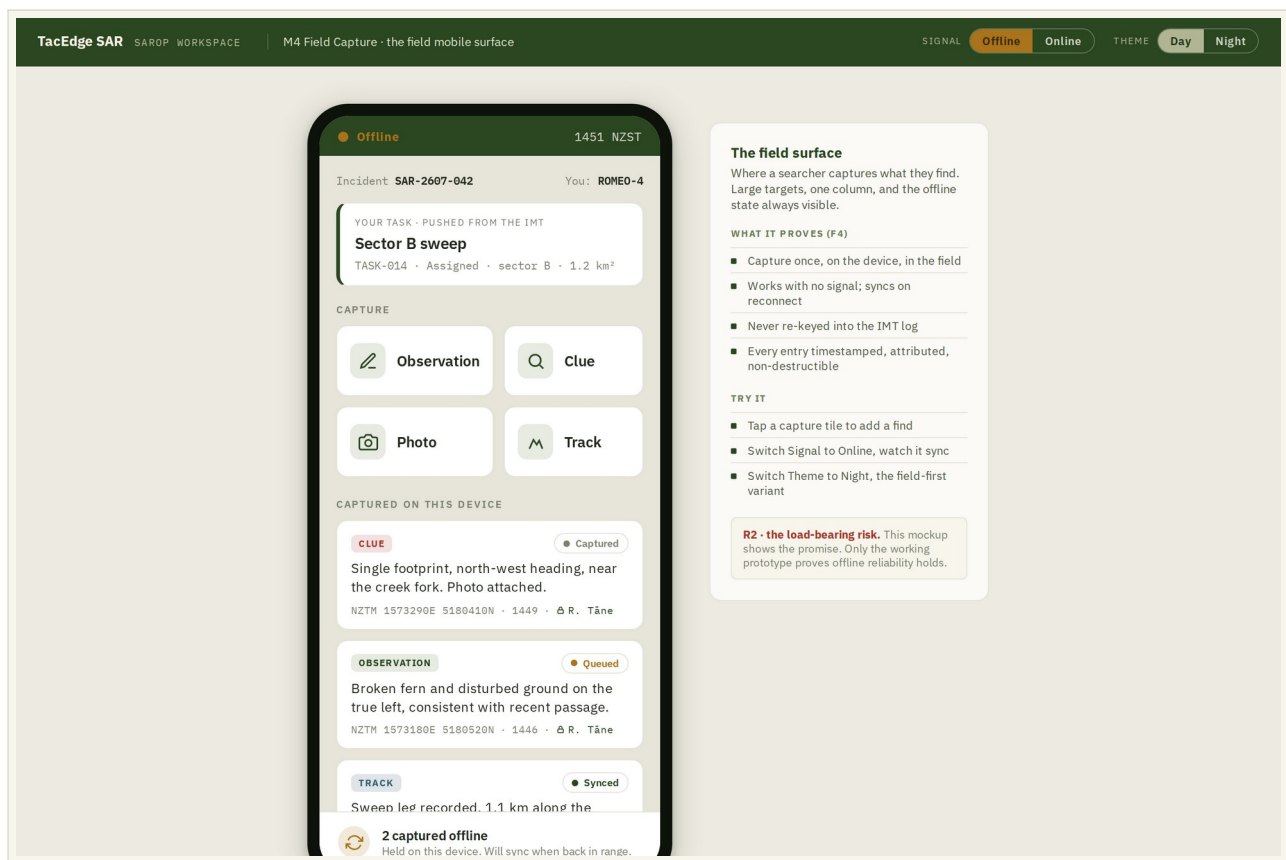
Getting a task to a field team in under a minute. The design commitment is one task object regardless of how the task is routed: direct to a controlled team on land, or for-affect to Coastguard on the water. The composer is built for speed, and the routing is the one thing that changes with the domain.

On the screen

- **Sub-minute stopwatch.** A timer starts the moment the composer is touched and reports the elapsed time on push, against the one-minute target.
- **The composer.** Subject, task type, one or more assigned resources with their sign-in state, attached geometry (area, path or single location), channel and push, top to bottom for speed.
- **The routing switch, the marine config made concrete.** On land the task is a direct tasking to the assigned team. Switch to marine and the same block becomes a task for-affect to Coastguard, which receives it as a parent and issues its own child tasks to its assets. Same object, one flow.
- **Push to the field.** Across app, SMS or email; the task lands on the board as Assigned and is logged.

M4 Field Capture

Flow F4 · offline capture Primary user: field SAR team Condition: offline, zero re-entry



M4 — the field surface on a phone, offline, with the sync state always visible.

Purpose

The field surface, on a searcher's phone. A team captures what they find once, in the field, often with no signal, and it must reach the incident log without being re-keyed. This is the surface that carries the offline reliability bet, and the design is field-first: large targets, one column, night mode as a first-class variant.

On the screen

- **Offline, always visible.** The signal state leads the screen and a persistent sync bar shows what is held on the device and waiting to send. Capture works with no signal.
- **Capture once.** Large tiles for an observation, clue, photo or track. Each capture is geo-referenced, timestamped, attributed and non-destructible from the moment it is made.
- **The sync lifecycle.** Captures move Captured to Queued to Synced; when signal returns they write to the record, and the screen shows nothing was re-keyed.
- **Night mode and an honest flag.** A first-class night variant for the surface that runs after dark, and an on-screen note that this mockup shows the promise of offline while only the working build proves it.

M5 Log & Records

The trusted record Written by the field, read by all Trusted on sight, reviewed remotely

TacEdge SAR SAROP WORKSPACE SAR-2687-042 **Active** Trusted record - timestamped, attributed, non-destructible

Log & records

Every action, decision, clue and communication, in one chronological record. Searchable during and after the operation, and reviewable remotely, with no district method to learn.

Search the record — subject, description, author, reference... fuzzy · during & after

All Decisions Clues Observations Comms Tasks

EVENT LOG 8 of 8 entries

- 1451 DECISION**
Sector priority
Prioritise sector A on hasty; hold sector B for daylight.
IC · LOG-0148
- 1449 CLUE**
Footprint near creek fork
Single footprint, north-west heading, near the creek fork. Photo attached.
R. Tāne · LOG-0142
- 1446 OBSERVATION**
Ground sign, true left
Broken fern and disturbed ground on the true left, consistent with recent passage.
R. Tāne · LOG-0141
- 1443 TASK**
TASK-014 Sector B sweep
Sector B sweep assignment.
IC · LOG-0138
- 1440 TRACK**
Sweep leg, sector B
Sweep leg recorded, 1.1 km along the sector B boundary.

LOG-0142 CLUE

Footprint near creek fork

Single footprint, north-west heading, near the creek fork. Photo attached.

PRIORITY **High**

STATUS **Active**

GEO-REFERENCE **NZTM 1573290E 5180410N**

ATTACHMENT **footprint.jpg**

Logged (you set) 1449

Recorded (system) **14:49:32 NZST**

Author **R. Tāne**

The recorded time and author are system-set and cannot be edited. The logged time can be adjusted for out-of-sequence capture.

HISTORY

- 1449 Captured in field (M4) R. Tāne

Edit Strike & correct Attach Print

RECORD STORE 4 files

M5 — the trusted record, with an entry struck and corrected, the original retained.

Purpose

The trusted record. Every action, decision, clue and communication in one chronological, searchable record that command, a reviewer or a coroner can trust on sight and review remotely, with no district-specific method to learn. It is the record the field captures into and the close-out reads from.

On the screen

- **One connected record.** The entries thread through every other surface: the stand-up decision, the task, the field captures, the comms, all in one timeline.
- **Dual timestamps.** An editable logged time for out-of-sequence capture, alongside a locked system-recorded time that cannot be changed. The immutable stamp is what makes the record trustworthy.
- **Struck and corrected, never deleted.** Corrections strike the original in place and keep it visible in the history. Nothing is lost, which is what makes remote review defensible.
- **Findable.** Fuzzy search during and after the operation, category filters, and the per-incident record store, every file date, time and user stamped.

M6 Forms Engine

Configurable forms Authored by the IMT Structured records, not PDFs

The screenshot displays the TacEdge SAR Forms Engine interface. The top header shows 'TacEdge SAR SAROP WORKSPACE' and 'SAR-2687-042' with an 'Active' status. The domain is set to 'LAND'. The left sidebar lists various forms under categories like 'SUBJECT & INTELLIGENCE', 'SEARCH URGENCY', 'PLANNING', 'FIELD & CASUALTY', and 'OUTWARD & DEBRIEF'. The main area is titled 'Search Urgency Assessment' and shows a 'FORMS-6' structured record. It includes a note about AI assistance: 'Values suggested from the missing-person intake. Confirm or adjust each factor. The score is assistance, it never decides for you.' The assessment factors and their scores are: Number of missing persons (2), Age (1), Medical conditions (1), Fitness (2), Experience (2), Reliability (2), Clothing (2), Equipment (2), and Weather (1). The total urgency score is 18, which is 'of 11-41'. A gauge indicates the score is in the 'most urgent' range. The response recommendation is 'Requires emergency response'. A note at the bottom states 'AI-assisted, human-verified. You set every value.'

M6 — the scored Search Urgency Assessment, banded live to an emergency response.

Purpose

The SAR form set as structured records rather than attached PDFs, and the place where land and marine configure most visibly. The centrepiece is the scored Search Urgency Assessment, which turns a set of factors into a banded response, with AI assistance that suggests values but never decides.

On the screen

- **Structured records.** The form set is navigable with completion status; a form is data on the record, sharing the log field set, not a document bolted on.
- **Scored urgency, banded live.** Each factor is scored and the total lands in a response band, emergency, measured or evaluate, with a gauge showing where it sits. A lower total means higher urgency.
- **The land / marine config pair.** Land uses the eleven-factor assessment; switch to marine and the whole factor set and banding change to the marine method. Same engine, different configuration.
- **AI-assisted, human-verified.** Values are suggested from the intake and marked as such; the coordinator confirms or adjusts every one. Concurrent editing lets more than one person work the same record.

M7 Notifications & Callout

Activation Coordinator initiates, members respond Every message on the record

TacEdge SAR SAROP WORKSPACE SAR-2607-042 Active One callout · every reply on the record

Notifications & callout

Call out the people you need from one list, across the channels they use, and watch replies land in one place. Every message sent and received is written to the record.

NEW CALLOUT app · SMS

6 recipients selected All ground Select all Clear

MESSAGE

SAR callout — SAR-2607-042. Overdue tramper, Kepler Track. Muster at Te Anau SAR base. Reply with availability and ETA.

App SMS Email Send callout

RESPONSES not sent

No callout sent yet.

Select recipients, then send. Replies appear here as they come in.

CONTACT ROSTER 10 contacts

All Ground Dog Water Medic

Agency

- ☒ **R. Tane**
Ground · Bushcraft · Nav
- ☒ **J. Mills**
Ground · Team leader
- ☒ **K. Rewi**
Dog · K9 · air-scent
- ☒ **S. Patel**
Medic · Advanced care
- ☒ **A. Hunt**
Ground · Nav · rope
- ☒ **M. Fox**
Water · Swiftwater
- ☐ **L. Chen**
Air · Air liaison
- ☐ **T. Wiremu**
Ground · Bushcraft
- ☐ **Coastguard**
Agency · Marine unit
- ☐ **LandSAR regional**
Agency · Coordination

COMMS RECORD feeds the log

Nothing sent yet. The callout and every reply will be recorded here and in the incident log.

M7 — one callout, with replies tracked live and written to the comms record.

Purpose

Activation. Calling out the people an operation needs from one list, across the channels they use, and tracking who is responding, replacing the fragmented scramble across separate messaging tools. Every message sent and received is written to the record.

On the screen

- **One roster.** Members and agencies with their capability, skills and on-call status, filterable, so recipients are chosen from a single list rather than assembled ad hoc.
- **Multi-channel callout.** The same message goes out across app, SMS and email, and the channel is recorded with it.
- **The live response board.** Recipients move from notified to responding with an ETA, or unavailable, with summary counts, so the coordinator can see who is actually coming and when.
- **On the record.** The callout and every reply are logged and feed the incident log, not fired into a channel that leaves no trace.

M8 Resources & Staging

Accountability & reconciliation Logistics and on-scene coordinator Everyone accounted for; hours reconciled

TacEdge SAR SAROP WORKSPACE SAR-2687-042 Active 3 in the field · accounted for

Resources & staging

Who is here, who is in the field, and who has signed out. Sign resources in on arrival and out on departure, so the operation always knows everyone is accounted for, and the deployed hours reconcile at close-out.

RESOURCE REGISTER 7 resources

RESOURCE	STATE	TASK	ON TASK	
IMT Command · 3 pax ICP · in 13:22	DEPLOYED	Command	1:30	Return
ROME0-2 Ground team · 4 pax Sector A · in 13:38	DEPLOYED	TASK-012	1:14	Return
ROME0-4 Ground team · 4 pax Sector B · in 13:52	DEPLOYED	TASK-014	1:00	Return
K9-1 Dog team · 1 dog Staging · in 14:05	AT STAGING	—	—	Deploy
MEDIC-1 Medic · S. Patel Staging · in 14:10	AT STAGING	—	—	Deploy
ROME0-1 Ground team · 4 pax Staging · in 13:15	RETURNED	TASK-009	1:45	Sign out
Heli-1 Air · Squirrel Wanaka	AVAILABLE	—	—	Sign in

RECONCILIATION 14:52

3 In the field
Every deployed team accounted for.

6 ON SCENE

2 AT STAGING

1 RETURNED

0 SIGNED OUT

Total deployed hours 5:29

Reconcile for close-out
Totals carry to M9 close-out as deployment hours and cost.

STAGING ready to deploy

K9-1
Dog team · 1 dog Deploy

MEDIC-1
Medic · S. Patel Deploy

M8 — the resource register, with the in-the-field accountability count and live deployed hours.

Purpose

Accountability and reconciliation. Always knowing who is in the field, signing resources in on arrival and out on departure, tracking deployed hours, and reconciling at close-out so nothing and no one is unaccounted for and the hours are ready for the cost record.

On the screen

- **Accountability first.** The count of teams in the field leads the screen, because the one thing an operation cannot lose track of is who is still out.
- **The sign-in ledger.** Every resource moves through a lifecycle, available, at staging, deployed, returned, signed out, with a single action to advance it.
- **Live deployed hours.** Time on task accumulates for deployed resources and totals for the operation, the raw material for close-out cost.
- **Staging and reconciliation.** A quick staging view of who is ready to deploy, and a reconciliation panel that carries the locked totals to close-out.

SAROP Workspace · Prototype Walkthrough · v0.1 · Internal · Pre-gate

10

M9 Close-out & Outward Record

Flow F5 · close-out Coordinator completes, command receives Zero re-entry; the confirmed outward record

TacEdge SAR SAROP WORKSPACE SAR-2607-042 Completed Zero re-entry · assembled from the record

Close-out

The whole operation, assembled from the record into one report. Nothing is re-keyed. Review, approve, and it becomes the confirmed outward record that feeds handover, debrief, the coroner, and the national picture.

Confirmed outward record locked · 2017 · IC A. Reid

Search & Rescue Operation Report

SAR-2607-042 Overdue tramper · Kepler Track Land · Category I Outcome: Subject found safe

Incident

Stood up 1422 Closed 2017

Duration 2h 25m Controller IC · A. Reid

Summary

Overdue tramper reported on the Kepler Track, last seen Luxmore Hut. Search urgency assessed as emergency response. Two ground teams tasked to sectors A and B; a footprint clue at 1449 narrowed the search north-west. Subject located safe in sector B at 1547 and walked out with ROMEO-4. Operation closed at 2017.

Timeline

1422 Stand-up, Category I, land, IC assigned.

1431 Weather: cloud lowering to 900 m after 1600.

1438 Search urgency assessed, emergency response.

1443 TASK-014 sector B sweep assigned (reassigned ROMEO-2 → ROMEO-4).

1449 Clue: footprint, NW heading near creek fork.

1547 Subject located safe, sector B.

Tasks & search coverage

Tasks issued 3 Area searched 2.1 km²

Sectors A, B Cumulative PoD 68%

TASK-012 Sector A TASK-014 Sector B TASK-015 Containment

Resources & deployed hours

ASSEMBLE

Incident & setup M1

Timeline & decisions M5

Tasks & coverage M3

Resources & hours M8

Forms completed M6

Regenerate

Assembled in 1.4s · 6 sources

Pulled from the live record. Nothing re-keyed.

OUTCOME

OPERATION OUTCOME

Subject found safe

APPROVE

Confirmed as outward record

Locked at 2017. Non-destructible from here.

OUTWARD RECORD FEEDS

Handover pack

next duty coordinator

Coroner-ready record

trusted, non-destructible

Team debrief

FORMS-21 / 22

M9 — the operation assembled from the record and confirmed as the outward record.

Purpose

The convergence surface. The whole operation is assembled from the record into one report, with nothing re-keyed, reviewed and approved by the Incident Controller, and confirmed as the authoritative outward record that feeds handover, the coroner, the team debrief and the national picture.

On the screen

- **Auto-assembly, zero re-entry.** The report is assembled from the record in seconds, pulling the incident, timeline, tasks, resources and hours, and forms. Each section is tagged with its source, because it is assembled, not authored.
- **The whole connected operation.** The report is this operation: the stand-up, the urgency assessment, the reassigned task, the footprint clue, the outcome and the reconciled hours, all threaded from the other surfaces.
- **Confirmed as the outward record.** Approval seals the report as locked, attributed and non-destructible, the trusted-record signature applied to the whole operation.
- **It feeds what comes next.** On confirmation the outward record is available for handover, coroner-ready review, team debrief and emission to the national picture, and the incident status is driven to Completed.

SAROP Workspace · Prototype Walkthrough · v0.1 · Internal · Pre-gate

11

The operation, end to end

Read in order, the nine surfaces walk a single operation from first call to closed record: stand up (M1), see the picture (M2), task a team (M3), capture from the field (M4), hold the trusted record (M5), score and complete the forms (M6), call out the crew (M7), account for the resources (M8), and close out to the confirmed outward record (M9). Because they share one record and one design system, they demonstrate as one product.

What is proven here, and what is next

This prototype tests desirability. It is the artefact for the conversation with a real SAR coordinator and a coordinating authority: is the coordinator-first design right, and are the flows fast and legible enough for a live job. That conversation is the next step, and it is what these screens are built for.

It does not test offline reliability. That is the load-bearing engineering risk, and only the working build proves it, with the field capture surface and its offline behaviour as the early, hard test. No amount of screen polish substitutes for that.

The through-line. One spine, many surfaces; land and marine as a configuration; a record that is trusted on sight and reviewable remotely; and an export boundary to a future national picture. The prototype makes those commitments visible so they can be tested before they are built.